Here are some examples of chemical formula.

Sodium Chloride (Nacl) : 1 atom of Sodium and 1 atom of chlorine

Ammonia (NH_3) : 1atom of Nitrogen and 3 atoms of Hydrogen

Glucose ($C_6H_{12}O_6$) : 6 Carbon atoms, 12 Hydrogen atoms and 6 Oxygen atoms

The chemical formula tells us the types of atoms and the number of each type of atom in one molecule of substance.

Table 3.5 Common compounds and their chemical formula

Names	Formula
Water	H_2O
Glucose	$C_6H_{12}O_6$
Salt	NaCl
Ethanol	C_2H_5OH
Ammonia	NH_3
Sulphuric Acid	H_2SO_4
Methane	CH_4
Sucrose	$C_{12}H_{22}O_{11}$

3.6 Atomicity

In chemistry, atomicity implies the total number of atoms present in one molecule of an element, compound or a substance. Let us see how to calculate the atomicity of elements. For example, oxygen exists as a diatomic molecule. It means that a molecule of oxygen contains two atoms hence its atomicity is 2.

$$O + O \longrightarrow O_2$$

Oxygen atom + Oxygen atom ⟶ Oxygen Molecule

Similarly a phosphorus molecule (P_4) contains 4 atoms and a sulphur molecule (S_8) contains 8 sulphur atoms. Hence, their atomicity is 4 and 8 respectively.

For molecule containing more than one types of atoms, simply count the number of each atom and that would be its atomicity. For example, one molecule of sulphuric acid (H_2SO_4) consists of 2 hydrogen atom, 1 sulphur atom and 4 oxygen atoms. Hence, its atomicity is 7(2+1+4).

One molecule of water (H_2O) contains two atoms of hydrogen and one atom of oxygen. Thus, the atomicity of water is three.

Table 3.6 Atomicity of some elements

Element	Atomicity	Elements	Atomicity
H	2	F	2
He	1	Ne	1
Li	1	Na	1
Be	1	Mg	**1**
N	2	P	4
O	2	S	8

ACTIVITY 4

Write down the atomicity of the following elements and compounds

Elements / Compounds	Atomicity
Cl	
Na	
K	
Ca	
H_2O	
Nacl	

3.7 Effect of temperature on Solid, Liquid and Gas

In solids, particles are arranged very closely. When solids are heated, the particles in them gain energy and vibrate vigorously. They move slightly further apart from one another. This causes the volume of matter to increase. This process is called expansion. How it happens?

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